2,8,7 2,8,8,1

Chem Test

Q1)

1) ~~Ac~~ 4
2) 1
3) 4
4) 1
5) 3
6) 4
7) 2
8) 1
9) 1
10) 4
11) 1
12) 2
13) 3
14) 1
15) 3

16) i) Refer table

a)
b)
c) Halogens
d) 13- Boron; 15—Nitrogen
e) Carbon

2,8,8,1

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IV

$$\frac{\overset{112}{\cancel{22.4}} \times 5}{\cancel{10}_2}$$

a) Reducing agent
b) lead
c) acidic
d)
e) F

iii)

i) Molar Mass $= 32 + 2(16)$
$\Rightarrow 32 + 32 \Rightarrow 64$

$\therefore$ No. of moles $= \frac{320^5}{64} \Rightarrow 5$ mole

~~$\therefore$ No. of~~
$\therefore$ Hence, $n = 5$
$\Rightarrow \frac{\text{Given Volume}}{22.4} = 5$

$\Rightarrow$ Given Volume $= 112g$

ii) When gases react, they do so in volume which bear a simple whole number ratio to one another & to the volume of product, provided the volume are measured in same condition of temperature and pressure.

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iii) $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
Molar Mass of propane = 44

44 g C_3H_8 requires 5×22.4 L of O_2
$\therefore$ 8.8 g of C_3H_8 = $\frac{5 \times 22.4 \times 8.8}{44}$

$\Rightarrow$ 22.4L

iv)

i) No. of moles = $\frac{\cancel{4.6}^{\,2}}{\cancel{23} \times 10 \cdot 5}$

$\Rightarrow$ 0.2 g-atoms

ii) Molar Mass = 64+32+16(4)+5(18)
$\Rightarrow$ 64+32+64+90
$\Rightarrow$ ~~100+96~~ $\Rightarrow$ 250g
$\Rightarrow$ ~~196g~~

$\therefore$ % of water of crystallisation = $\frac{5 \times 18}{\cancel{250}_{\,5}} \times \cancel{100}^{\,2} \Rightarrow$ 36%

iii) VD = Emperical Weight
Now, MW = 2VD
$\therefore n = \frac{MW}{EFW} \Rightarrow \frac{2VD}{EFW} \Rightarrow 2$

$\therefore$ Molecular Formula = $[XY_2]_2 \Rightarrow X_2Y_4$

v)
i) Phosphorus (At no: 15)
ii) Potassium (At no: 19)
iii) Oxygen
iv) Berelliyum
v) Helium

Section-B

1)
i) No. of moles = 1
$\Rightarrow \frac{\text{Given Mass}}{12} = 1$
$\Rightarrow$ Given Mass = 12g

ii) a)
1) pH decreases
2) pH increases

b) At no: $15 \rightarrow 2,8,5$
$\therefore$ Group 15

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iii) $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$
$8.2g$

a) No. of moles of $Ca(NO_3)_2 = \frac{8.2}{164} \Rightarrow \underline{\underline{0.05 \text{ moles}}}$

Since 2 moles of $Ca(NO_3)_2$ produce 4 moles of NO_2
$\therefore$ 1 " " 1/2 " "

$\therefore NO_2$ has 2 moles

$\therefore n = 2$
$\Rightarrow \frac{\text{Given Vol.}}{22.4} = 2$
$\Rightarrow$ Given Volume $= \underline{\underline{44.8 L}}$

b) Molar Mass of Calcium Nitrate $= 164g$

Now, $(2 \times 164g)$ of $Ca(NO_3)_2$ produces (2×56) CaO

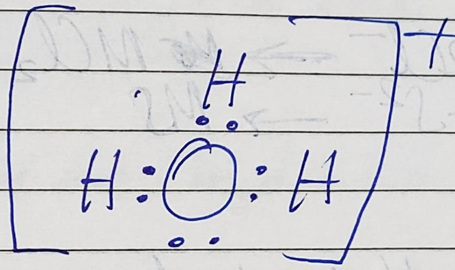
$\therefore$ CaO produced $= \frac{\cancel{2} \times 164}{\cancel{2} \times 56} \Rightarrow \underline{\underline{2.92\ g}}$

iv) i) Give Reason

a) Zinc produces a gelatinous white ppt when reacting with conc. NaOH whereas Aluminium produces white ppt

b) Calcium Nitrate produces no ppt on reaction whereas Lead Nitrate produces white ppt

ii)



2) a)

b)

ii) No. of moles $= \frac{21}{42} \Rightarrow 0.5$ moles

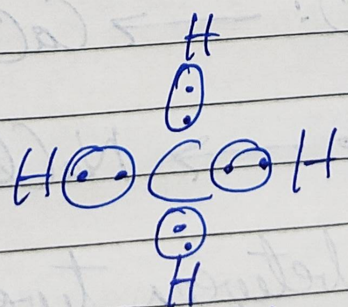
iii)

a) X has low oxidising power than Y.

b) X ~~ko~~ is more electronegative

c) X is placed to the right of Y

iv) i)



ii) Group 2 elements are also called Alkaline earth metals

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c)

b) They are metallic in nature

3) i) $M^{2+} + 2Cl^- \longrightarrow MCl_2$

$M^{2+} + S^{2-} \longrightarrow MS$

ii)

a) Because H has 1 valence e^- to pair, whereas O needs 2 e^- to attain stability.

b) Because Cl needs 1 e^- to stabilize whereas C need 4 e^-.

iii) i) $Mg: + \cdot Cl \ \cdot Cl \longrightarrow Mg^{2+} + Cl_2^-$

ii) $Ca: + O: \longrightarrow CaO$

iii) $Na\cdot + Cl\cdot \longrightarrow NaCl$

iv) A bond formed between two atoms by pair of electrons, provided entirely by one of the combining atoms. Condition are:

•) One of the atoms must have atleast 1 pair of electron

•) The other atom must have incomplete ~~pair~~ number of electron

Examples are : Hydronium ion;

4) i)

a) 2, 8, 8, 2

b) Period-~~3~~4; Group-2

ii)

a) +2

b) Electrovalent Compound

iii) Under Constant temperature & Pressure, Volume of gas is equivalent to number of moles of that gas.

•) Used to determine atomicity of gases.)

•) forms a relationship between molar mass and VD ~~of~~ a gas.

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iv) Refer table

i) A and B; C and D

ii) A and C; B and D

iii) D

5) i) $CaCO_3 \longrightarrow CaO + CO_2$

mass of $CaO = 14g$

MM of $CaO = 40+16 \Rightarrow \underline{\underline{56g}}$

$\therefore n = \frac{\cancel{14}^{1}}{\cancel{56}_{4}} \Rightarrow n = \underline{\underline{0.25 \text{ moles}}}$

Now, MM of $CaCO_3 = 100g$

$\therefore x = 0.25 \times 100 \Rightarrow \underline{\underline{25g}}$

ii) MM of $CO_2 = 12+32 \Rightarrow 44g$

iii)

Fe + 2HCl

iV) i) $ZnCO_3$, CuO

ii) Normal : Na_2SO_4 ; Acidic : $NaHSO_4$

iii) ~~Pb~~ Lead Chloride

6) i)

a) $CuSO_4 . 5H_2O$

b) NaCl

ii) Salts that spontaniously lose their water of crystallization is called Efflorescence.

The extreme form of taking water from atmosphere is called Deliquesent

iii) .) Turns blue litmus red

.) React with metal to form to form a salt and liberate hydrogen

.) React with base to form Normal salt & H_2O

iv) a) Acidic
b) Basic
c) Basic
d) Basic
e) Acidic
f) Acidic